

Data Analyst Interview Case Practice Set

Python + Pandas | Data Audit, Cleaning, Filtering, Conditional KPIs

Candidate Scope

These cases are designed for students who have completed Python basics and Pandas foundations up to data inspection, cleaning, type conversion, missing value handling, duplicate checks, filtering, conditional columns, gross vs valid metrics, and basic business summaries.

Do not use dashboards, charts, machine learning models, statistical testing, joins-heavy modelling, or predictive analytics unless the interviewer separately asks for it. The expected work is a clean notebook with a business-readable explanation of the assumptions, filters, exclusions, and final metrics.

Most datasets are hosted on Kaggle or public repositories. Kaggle datasets may require sign-in to download.

General Candidate Instructions

1. Read the problem as a business request, not as a list of Pandas commands.
2. Begin with a short data audit: shape, columns, data types, missing values, duplicates, and suspicious values.
3. Create business-safe metrics only after cleaning and validating the required columns.
4. When excluding or flagging records, explain the business reason. Do not silently delete data.
5. Where a dataset has multiple possible interpretations, state the assumption used in your analysis.
6. Submit the notebook and a short business note. The business note should explain what changed between raw and corrected numbers.

1. Retail Revenue Reconciliation and Invoice Quality Audit

Dataset: Online Retail - UCI Machine Learning Repository

<https://archive.ics.uci.edu/dataset/352/online+retail>

Interview Prompt

A UK-based online retailer reports that monthly revenue in its internal report is higher than the finance team's number. You are given transaction-level invoice data containing invoice numbers, stock codes, descriptions, quantities, invoice dates, unit prices, customer IDs, and countries. Investigate why the revenue number may be inflated or unreliable. Build a clean analysis-ready view of the transactions, identify cancelled or negative-quantity records, missing customer IDs, suspicious unit prices, duplicate-like invoice lines, and country-level inconsistencies. Your final response should present the corrected revenue logic, the difference between raw and valid revenue, the records excluded from reporting, and the business reason behind each exclusion.

Expected Data Issues and Edge Cases

Expect invoice cancellations, negative quantities, zero or unusual prices, missing customer identifiers, repeated invoice lines, and countries with small but unusual order patterns.

Allowed Scope

Use Pandas inspection, type checks, missing-value handling, duplicate checks, filtering, conditional revenue columns, and basic country/product/customer summaries. Do not build visualizations or predictive models.

2. Superstore Profit Leakage Investigation

Dataset: Superstore Dataset - Kaggle

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Interview Prompt

A regional sales head says total sales look healthy, but profit is not improving. You are given superstore order data with customer, segment, region, category, sub-category, sales, quantity, discount, profit, order date, and ship date. Investigate whether the business has a discount-driven profit leakage problem. Identify orders where high sales still resulted in negative profit, categories or sub-categories where discounts are repeatedly connected with losses, suspicious rows where profit or sales do not match the business expectation, and regions where the issue is more visible. Your final response should separate high-revenue business from genuinely profitable business and explain which records should be reviewed by the pricing or sales team.

Expected Data Issues and Edge Cases

Expect high-discount orders, high-sales low-profit rows, negative-profit transactions, inconsistent shipment delays, and categories where volume hides profitability problems.

Allowed Scope

Use cleaning, date conversion, filtering, calculated columns, conditional flags, and basic grouped summaries. No dashboarding, charts, or advanced statistics required.

3. Hotel Booking Cancellation and Revenue-at-Risk Audit

Dataset: Hotel Booking Demand - Kaggle

<https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

Interview Prompt

A hotel revenue manager says the booking pipeline looks strong, but actual occupancy is weaker than expected because cancellations are high. You are given booking records for city and resort hotels with cancellation status, lead time, stay nights, guest counts, meal type, country, market segment, deposit type, booking changes, ADR, and special requests. Investigate which bookings should be treated as real revenue opportunities and which should be treated as cancellation risk or data-quality issues. Identify cancelled high-value bookings, zero-guest or impossible-stay records, missing country values, long-lead bookings with high cancellation exposure, and hotel or segment combinations that need attention. Your final response should explain how you would define potential revenue, valid stayed revenue, and revenue at risk using only the available columns.

Expected Data Issues and Edge Cases

Expect cancelled bookings, zero adults/children/babies, missing country values, high lead-time cancellation risk, unusual ADR values, and hotel/market-segment differences.

Allowed Scope

Use Pandas cleaning, filtering, date/number checks where applicable, conditional revenue logic, suspicious-record flags, and basic summaries. No ML cancellation prediction or visualization.

4. Food Delivery Cost and Profitability Reconciliation

Dataset: Food Delivery Cost and Profitability - Kaggle

<https://www.kaggle.com/datasets/romanniki/food-delivery-cost-and-profitability>

Interview Prompt

A food delivery platform says order volume is growing, but profitability is unclear. You are given order-level data with customer IDs, restaurant IDs, order value, delivery fee, payment method, discounts/offers, commission fee, payment processing fee, refunds or chargebacks, and timestamps. Investigate whether the platform is losing money because of discounts, refunds, low commission, high processing cost, or order-level edge cases. Define raw order value, valid collected revenue, platform cost, and net contribution using business rules that you clearly state. Identify orders where revenue exists but platform contribution is negative, discount-heavy orders that failed to create profit, refund or chargeback rows, and payment methods that create higher risk.

Expected Data Issues and Edge Cases

Expect discounts, commission and payment processing fees, refunds or chargebacks, multiple payment methods, order values that do not directly equal platform profit, and possible timestamp issues.

Allowed Scope

Use data audit, calculated columns, conditional valid revenue/profit logic, filters, and basic order/payment/restaurant summaries. No charts or profitability model required.

5. NYC Taxi Fare and Suspicious Trip Audit

Dataset: NYC Yellow Taxi Trip Data - Kaggle

<https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data>

Interview Prompt

A city transportation analytics team wants to verify whether taxi fare reports are reliable before publishing weekly operating metrics. You are given trip-level taxi data with pickup/drop-off timestamps, passenger count, trip distance, fare components, total amount, payment type, vendor details, and location IDs. Investigate suspicious trips and clean the dataset enough to produce trustworthy trip and fare metrics. Identify zero-distance trips with positive fare, zero-passenger or unusually high-passenger trips, negative or unrealistic fare rows, very short trips with high total amount, cash vs card payment differences, and records where pickup time is after drop-off time. Your final response should explain which trips are valid for reporting and which trips should be excluded or flagged.

Expected Data Issues and Edge Cases

Expect timestamp issues, zero or extreme passenger counts, zero distance, unusual fares, negative values, fare-component mismatches, and payment-type differences.

Allowed Scope

Use Pandas datetime conversion, filtering, conditional flags, basic fare calculations, and simple summaries. No maps, geospatial analysis, visualization, or forecasting.

6. Bank Marketing Campaign Conversion Audit

Dataset: Bank Marketing - UCI Machine Learning Repository

<https://archive.ics.uci.edu/dataset/222/bank+marketing>

Interview Prompt

A bank's marketing team ran phone campaigns to sell term deposits. Management wants to understand whether the campaign is efficient or whether the team is spending too much effort on low-conversion segments. You are given customer and campaign data including age, job, marital status, education, default, balance, housing loan, personal loan, contact type, campaign contacts, previous campaign results, duration, and final subscription result. Investigate campaign quality using only descriptive analysis. Identify segments with high contact effort but low subscription, records with unknown customer attributes, clients repeatedly contacted without conversion, and campaign conditions where successful subscriptions appear more common. Your final response should recommend which campaign records or customer groups deserve review before the next campaign round.

Expected Data Issues and Edge Cases

Expect unknown category values, repeated contacts, no previous contact indicators, duration-driven interpretation risk, loan/default flags, and subscription imbalance.

Allowed Scope

Use cleaning, category standardization, filtering, conditional flags, value counts, and basic conversion summaries. Do not build a prediction model.

7. Telco Customer Churn Revenue Exposure Review

Dataset: Telco Customer Churn - Kaggle

<https://www.kaggle.com/datasets/blatchar/telco-customer-churn>

Interview Prompt

A telecom company wants to understand which customer groups are causing revenue exposure through churn. You are given customer-level data with tenure, contract type, payment method, paperless billing, monthly charges, total charges, subscribed services, demographic fields, and churn status. Investigate whether churn is concentrated among specific billing, contract, service, or charge patterns. Before analysis, validate numeric fields such as total charges and detect blank or invalid values. Identify high monthly-charge customers who churned, month-to-month customers with higher churn exposure, payment methods connected with churn, and customers with short tenure but high monthly charges. Your final response should separate customer count from revenue exposure so management can prioritize retention action.

Expected Data Issues and Edge Cases

Expect numeric columns stored as text, blank total charges, categorical service fields, high monthly charge churners, short-tenure churners, and contract/payment segment differences.

Allowed Scope

Use Pandas cleaning, type conversion, filters, conditional flags, and basic group-level churn summaries. No model training or churn prediction.

8. Online Course Engagement and Completion Quality Check

Dataset: Online Course Student Engagement Metrics - Kaggle

<https://www.kaggle.com/datasets/thedevastator/online-course-student-engagement-metrics>

Interview Prompt

An online learning platform says course enrollments are increasing, but completion quality is not improving. You are given student-course level records with engagement, activity, progress, performance, enrollment, and completion-related fields. Investigate whether the platform is attracting students who enroll but do not complete, whether some course categories have weak completion, and whether high engagement always results in completion. Clean and validate missing or inconsistent student/course fields, create clear definitions for active learners, completed learners, inactive learners, and at-risk learners, and identify records where the data looks contradictory. Your final response should help the academic operations team decide where learner support is needed.

Expected Data Issues and Edge Cases

Expect missing engagement fields, low activity with completed status, high activity with incomplete status, course/category differences, and possible fee or enrollment inconsistencies depending on the file version.

Allowed Scope

Use inspection, missing-value decisions, conditional learner flags, filtering, and basic completion summaries. No dashboards, model training, or visualizations.

9. Retail Sales, Returns, and Shipping Reconciliation

Dataset: Retail Sales, Returns & Shipping Dataset - Kaggle

<https://www.kaggle.com/datasets/kunalmalviya06/retail-sales-returns-and-shipping-dataset>

Interview Prompt

A retail operations team says net sales are inconsistent across sales, returns, and shipping reports. You are given a retail dataset combining sales transactions, return records, and shipping cost details. Investigate why gross sales may not match net business value. Create a defensible way to separate gross sales, returned value, valid net sales, shipping cost exposure, and transactions that require manual review. Identify high-value returned orders, products or categories with repeated returns, shipping-heavy orders, suspicious rows where return or shipping values do not align with sales, and records that could distort reporting if included blindly.

Expected Data Issues and Edge Cases

Expect return flags or return amounts, shipping costs, sales values, possible duplicate-like transactions, high shipping costs, returned orders, and product/category-level issues.

Allowed Scope

Use data audit, filtering, conditional net-sales logic, duplicate checks, missing-value handling, and basic product/category summaries. No visualization or advanced profitability modelling.

10. D2C E-Commerce Funnel Drop-Off Investigation

Dataset: Direct-to-Consumer E-Commerce Funnel Dataset - Kaggle

<https://www.kaggle.com/datasets/yashch05/direct-to-consumer-e-commerce-funnel-dataset>

Interview Prompt

A D2C brand says website visits are strong, but completed purchases are not growing. You are given session or funnel-level data that tracks user movement from visit to product view, cart, checkout, and purchase, along with campaign, device, and order-related fields. Investigate where users are dropping off and whether the issue is tied to device type, campaign source, cart behavior, discount usage, payment failure, or suspicious sessions. Define valid sessions, product-view sessions, cart sessions, checkout sessions, and purchase sessions using the available data. Identify sessions with checkout but no purchase, cart abandonment patterns, campaigns that bring traffic without conversion, and any records that should not be counted as valid conversion events.

Expected Data Issues and Edge Cases

Expect multi-stage funnel fields, missing stage values, sessions without purchase, campaign/device differences, possible zero-value purchases, discount sessions that do not convert, and duplicate or repeated user sessions.

Allowed Scope

Use filtering, conditional flags, value counts, basic conversion-rate calculations, and segment summaries. No visualization, A/B testing statistics, or predictive modeling.